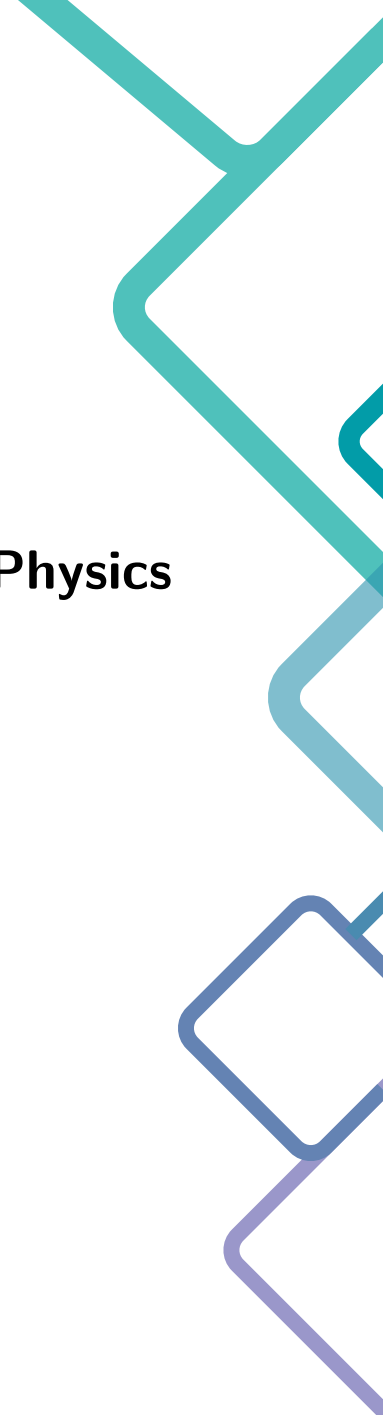




Lecture Notes

Geometrical Anatomy of Theoretical Physics

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Structure of this Course

In theoretical physics we mainly deal with three big part of mathematics, namely, ‘Analysis’, ‘Algebra’ and ‘Geometry’. And at the mutual intersection of these three, we have different branches of physics like ‘Quantum Mechanics’, ‘General Relativity’, ‘Statistical Mechanics’, etc. see [fig. 0.1](#).

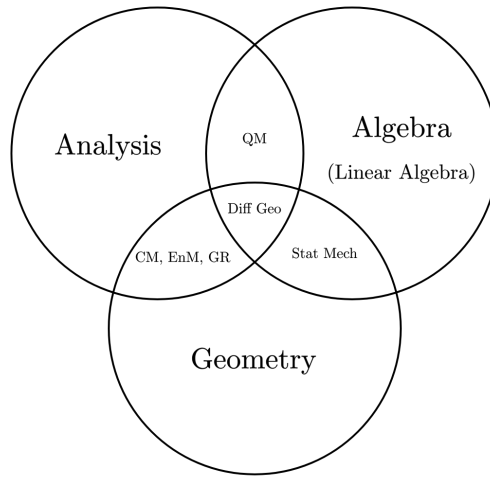


Figure 0.1: Structure of Theoretical Physics

This course mainly focuses on differential geometry and topology, and their applications in theoretical physics. And we will start with the basic propositional logic and set theory, and then move to the topology and geometry of manifolds, and then we will see the applications of these in physics. Following is the structure of this course:

- Logic
- Set Theory
- Topology
- Topological Manifolds
- Differential Manifolds
- Bundles
- Geometry: Symplectic Geometry, Metric Geometry, etc.
- Physics: Classical Mechanics, Electrodynamics, Quantum Mechanics, Statistical Mechanics, Special and General Relativity, etc.

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1 Differentiable Manifolds

LECTURE 12

§1.1 Differential Forms and de Rham Cohomology

Definition 1.1 (Differential n -form):

Let M be a smooth manifold, and $0 \leq n \leq \dim M$. A differential n -form on M is $(0, n)$ -tensor field ω that is totally antisymmetric, i.e.,

$$\omega(X_1, \dots, X_n) = \text{sgn}(\sigma) \omega(X_{\sigma(1)}, \dots, X_{\sigma(n)})$$

where $\sigma \in S_n = \text{Perm}(n)$ is a permutation of n elements, $\text{sgn}(\sigma)$ is the sign of the permutation, and $X_1, \dots, X_n \in \Gamma(\mathcal{T}M)$ are vector fields on M .

Let's look at some examples of differential forms.

Example 1.2

- a) A manifold M is *orientable* if it admits an oriented atlas, i.e., an atlas $\{(U_\alpha, x_\alpha)\}$ in which all the transition maps $x_\beta \circ x_\alpha^{-1} : x_\alpha(U_\alpha \cap U_\beta) \rightarrow x_\beta(U_\alpha \cap U_\beta)$ have positive determinant.

If M is orientable, then there exists a nowhere vanishing *top-form* ($n = \dim M$) ω on M , providing a volume.

- b) The electromagnetic field strength F in electrodynamics is a differential 2-form.

- c) In classical mechanics, define the configuration space Q of a system as the space of all possible configurations of the system, i.e., the space of all possible values of the generalized coordinates. Then the phase space of the system is the cotangent bundle $\mathcal{T}^*Q$ of the configuration space.

On phase space, we have a canonical 2-form ω called the *symplectic form*. To define it, we need some more machinery, but the anti-symmetry of ω is the reason why we have the minus sign in Hamilton's equations of motion.

Define the set of all differential n -forms on M as $\Omega^n(M)$, which by definition is a $\mathcal{C}^\infty(M)$ -module. Generally, taking a tensor product of a form doesn't give us a form, but we want to be able to multiply forms together to get one, so we define the *wedge product*.

Definition 1.3 (Wedge Product):

Let $\omega \in \Omega^n(M)$ and $\eta \in \Omega^m(M)$ be two differential forms on M . The wedge product of ω and η is a differential $(n+m)$ -form defined as

$$\begin{aligned} (\omega \wedge \eta)(X_1, \dots, X_{n+m}) &= \frac{1}{n!} \frac{1}{m!} \sum_{\sigma \in S_{n+m}} \text{sgn}(\sigma) (\omega \otimes \eta)(X_{\sigma(1)}, \dots, X_{\sigma(n+m)}) \\ &= \frac{1}{n!} \frac{1}{m!} \sum_{\sigma \in S_{n+m}} \text{sgn}(\sigma) \omega(X_{\sigma(1)}, \dots, X_{\sigma(n)}) \eta(X_{\sigma(n+1)}, \dots, X_{\sigma(n+m)}). \end{aligned}$$

Simplest example of wedge product is take two 1-forms ω and η , then

$$(\omega \wedge \eta)(X_1, X_2) = \sum_{\sigma \in S_2} \text{sgn}(\sigma) \omega(X_{\sigma(1)}) \eta(X_{\sigma(2)}) = \omega(X_1) \eta(X_2) - \omega(X_2) \eta(X_1).$$

Thus, we can write $\omega \wedge \eta = \omega \otimes \eta - \eta \otimes \omega$.

Recall that given a smooth map $\phi : M \rightarrow N$ between two smooth manifolds, we induce pull-backs $\phi_p^* : \mathcal{T}_{\phi(p)}^* N \rightarrow \mathcal{T}_p^* M$ on the cotangent spaces for each $p \in M$. With this global pull-back map $\phi^* : \Gamma(\mathcal{T}^* N) \equiv \Omega^1(N) \rightarrow \Gamma(\mathcal{T}^* M) \equiv \Omega^1(M)$ is defined pointwise.

Definition 1.4 (Pull-back of Forms):

Let $\phi : M \rightarrow N$ be a smooth map between two smooth manifolds, and let $\omega \in \Omega^n(N)$ be a differential n -form on N . The pull-back of ω by ϕ is a differential n -form on M defined as

$$(\phi^*\omega)(X_1, \dots, X_n) = \omega(\phi_*(X_1), \dots, \phi_*(X_n))$$

where $\phi_* : \mathcal{T}M \rightarrow \mathcal{T}N$ is the (global) push-forward map induced by ϕ .

This can be equivalently defined pointwise as $(\phi^*\omega)_p = \phi_{\phi(p)}^* \omega_{\phi(p)}$ for each $p \in M$.